# Chapter 7

# Branch-and-Price for Integrated Aircraft Routing and Maintenance

## 7.1 Purpose and status of the proposed approach

The compact formulation assigns flights and maintenance decisions directly to aircraft. Chapter 6 reduces its size and strengthens some of its inequalities. A complementary approach is to optimize over complete integer scheduling patterns for individual tails, generating those patterns only when they can improve a coordinating master problem. This chapter develops that approach for the full A/B/C/D maintenance hierarchy through Dantzig–Wolfe decomposition and branch-and-price.

*This chapter is a methodological proposal submitted for review before implementation. No branch-and-price solver, measured speedup, generated root bound, or computational optimality result is claimed here. The results in Chapter 9 concern the previously implemented methods.*

The target is an exact reformulation of the selected compact model, not a new interpretation of maintenance legality. The initial proposal retains the aircraft-indexed binary trigger formulation, hard flight coverage, full ABCD checks, and the same assignment domains, initial conditions, hierarchy, maintenance durations, deferral policy, cost accounting and routing flags. The optional global-trigger formulation is outside this initial decomposition because its cross-aircraft variables require a separate separability analysis. In particular, replacing a compact constraint by a stricter physical replay rule would change the comparison problem and is not part of this proposal.

For an unambiguous reference, the equivalence claim in this chapter concerns the default full-maintenance profile of `MILP_Sheduler`: hard coverage; aircraft-indexed sparse $z$ variables; day-specific $y$ and $\gamma$ bounds; C2–C3 and interval-clique overlap enabled; capacity, hierarchy, C12, C12b, C13b and corrected split C13 enabled; three-day A/B deferral; global $M_{\mathrm{BIG}} = 9{,}999{,}999$ for C13/C13b; and the default weak C9 link.

The C14b duration constant is $M_{\mathrm{C14b}} = \max_c K_c^{\mathrm{dur}} + 1$. The global-trigger, soft-coverage, no-capacity, reachability, tight-C13-$M$, original-paper-C13, strong-link, strong-conflict and sanity-fixing options are disabled. The documented compatibility C15 component is empty; C8 supplies the maintenance-flight blocking rows. Any experiment with another option set requires the corresponding local set and coefficient definitions to change.

## 7.2 Literature review and positioning

### 7.2.1 Decomposition and exact integer optimization

Dantzig and Wolfe [11] distinguish constraints belonging to individual subproblems from constraints connecting those subproblems. Their coordinating linear program uses columns proposed by the subproblems, while its dual prices define successive subproblem objectives. The relevance here is the separation between a tail's routing and maintenance decisions and the fleet-wide demands for flight coverage and maintenance-station capacity.

Decomposing an integer problem requires more than solving the coordinating LP. Barnhart et al. [3] describe column generation within a branch-and-bound tree, including pricing at tree nodes and branching rules compatible with implicit columns. This distinction is essential: column generation at the root yields a relaxation bound, whereas exact integer optimization requires pricing under the restrictions imposed throughout the search tree. An integer solve over a fixed, generated column pool can produce an incumbent, but does not certify the optimum over omitted columns.

### 7.2.2 Airline strings, routing and maintenance

Barnhart et al. [2] formulate aircraft fleeting and routing with flight strings, allowing maintenance and connection requirements to be represented within a route-based description. That work motivates replacing individual assignment decisions by structured aircraft duties. It is distinct from the general branch-and-price paper by Barnhart et al. [3], although both were published in 1998.

Desaulniers et al. [14] develop daily routing models using set partitioning and time-constrained networks, with column generation and compatible branching. Their profit-maximization setting must not be confused with the minimization problem in this thesis: the direction of a relaxation bound depends on the objective sense. Their methodological relevance is the connection between decomposed routing formulations and branching decisions that remain expressible in pricing.

Sarac et al. [35] specifically address operational aircraft maintenance routing through branch-and-price. This provides direct application precedent for generating maintenance-aware aircraft patterns. It does not by itself establish that their maintenance conventions

equal the ABCD hierarchy, initial-state treatment and station-capacity accounting used here. Those model-specific details must be stated and validated independently.

The compact model of Khaled et al. [23] remains the formulation reference for the thesis, together with the modifications in Chapters 4 and 5. The proposed contribution is therefore not the invention of branch-and-price. It is a precisely defined full-ABCD decomposition of that reference model, with reconstructable pattern certificates, transparent lower-bound certification and a controlled evaluation against the compact formulation.

Table 7.1: Distinctions between route-based solution procedures.

| Procedure | Columns considered | What can be certified |
|---|---|---|
| Complete pattern enumeration | Every feasible local integer pattern | Full master LP and integer optima, if solved |
| Root column generation | Improving patterns generated at the root | Full master LP bound after exact pricing certification |
| Restricted integer master | A fixed generated pool | A feasible incumbent; not generally full integer optimality |
| Branch-and-price | Patterns generated under restrictions at every node | Integer optimality if the complete search is certified |

## 7.3 Tail-local integer scheduling patterns

Flights, tails, stations, days and check types retain the sets $\mathcal{F}$, $\mathcal{P}$, $\mathcal{A}$, $\mathcal{D}$ and $\mathcal{C} = \{A, B, C, D\}$, respectively. Let $\mathcal{K} \subseteq \mathcal{A}$ be the maintenance stations. For each tail $j$, collect all of its original binary decisions into a vector $v_j$. Its components include $x_{ij}$, $z_{ijdc}$, $y_{jdc}$, $\gamma_{jdc}$ and maintenance-start auxiliaries. The mathematical symbol $\gamma_{jdc}$ corresponds to `mega[j,d,c]` in the implementation. Variables absent from a sparse domain are represented as zero when coefficients are evaluated.

The selected compact model's local feasible set is

$$\mathcal{Q}_j = \left\{ v_j \in \{0,1\}^{n_j} : A_j v_j \leq h_j,\ E_j v_j = e_j,\ \ell_j^{\text{var}} \leq v_j \leq u_j^{\text{var}} \right\}. \tag{7.1}$$

The matrices denote exactly the previously defined tail-local constraints; they do not introduce a replacement maintenance model. Table 7.2 specifies their ownership. Flight coverage C1 and shared capacity C10 are excluded from $\mathcal{Q}_j$ and retained in the master.

The table is the decomposition claim, row by row: every nonempty original row other than C1 and C10 is copied without changing its index sets, coefficients, sense or bounds into exactly one tail block. For example, C9 uses the implementation's default weak link: for C/D checks it links the assignment to the first candidate trigger day, not the stronger sum-over-days link. The sparse $Z$ domain, impossible-state upper bounds

Table 7.2: Constraint ownership in the proposed decomposition.

| Constraint group | Owner | Meaning |
|---|---|---|
| C1 | Master | Hard exact coverage of every flight |
| C10 | Master | Shared station/day capacity across tails and check types |
| C2–C3 | Tail pricing | Per-aircraft equipment flow and minimum turn |
| Overlap | Tail pricing | Per-aircraft temporal clique rows |
| C8 | Tail pricing | Same-day blocking and deferred-trigger escape blocking |
| C9 | Tail pricing | Default weak trigger/assignment link, exactly as coded |
| C11 | Tail pricing | Per-tail/day/check sum of triggers equals $y_{jdc}$ |
| Hierarchy | Tail pricing | $\gamma$ equalities from the selected checks |
| C12, C12b | Tail pricing | Calendar spacing and initial calendar deadline |
| C13, C13b | Tail pricing | Split hour rows and pre-horizon accumulated hours |
| C14, C14b | Tail pricing | At most one check/day and multi-day duration |
| C15 | No rows | Empty compatibility component; C8 performs blocking |
| Start links | Tail pricing | $s_{jdc}$ detects $y_{jdc}$ starts for objective cost |
| Variable domains | Tail pricing | Sparse $X, Z$, binary domains and day-specific zero bounds |
| Objective | Tail pricing | Assignment cost plus maintenance-start penalty |

and hierarchy bounds are also part of that block, not merely optional presolve. C15 has no active rows, so it contributes nothing to either block.

For clarity, let $H^c_{jdd'}$ be the assigned flight-minute accumulation in a C13 interval, let $Y^c_{jdd'}$ be the sum of intervening hierarchy reset indicators, and let $T_c = 60H_c$ be the flight-hour threshold in minutes when the input threshold $H_c$ is expressed in hours. Under the selected default profile, C13 uses the single global constant $M_{\text{BIG}} = 9{,}999{,}999$, not the optional interval-specific tightening. The reference split rows are

$$H^c_{jdd'} = \sum_{i \in \mathcal{F}_{d,d'} : (i,j) \in X} t_i x_{ij}, \tag{7.2}$$

$$Y^c_{jdd'} = \sum_{r \in \mathcal{D}:\ d<r<d'} \gamma_{jrc}. \tag{7.3}$$

where $\mathcal{F}_{d,d'}$ uses the same departure-day interval convention as the compact builder. The enforced pair is

$$H^c_{jdd'} \le T_c + M_{\text{BIG}} Y^c_{jdd'} + M_{\text{BIG}} \gamma_{jdc}, \tag{7.4}$$

$$H^c_{jdd'} \le T_c + M_{\text{BIG}} Y^c_{jdd'} + M_{\text{BIG}} \gamma_{jd'c}. \tag{7.5}$$

All terms use minutes. For each endpoint day $d'$, C13b is the additional opening-horizon

row

$$H^{c,0}_{jd'} = \sum_{i \in \mathcal{F}_{0,d'}:(i,j)\in X} t_i x_{ij}, \tag{7.6}$$

$$H^{c,0}_{jd'} \le 60(H_c - h^0_{jc}) + M_{\text{BIG}} \left( \sum_{r \in \mathcal{D}: r<d'} \gamma_{jrc} + \gamma_{jd'c} \right), \tag{7.7}$$

where $h^0_{jc}$ is the input's initial accumulated A/B flight-hour exposure. The interval domains and opening-horizon rows are inherited from the reference model. Branch-and-price does not, by itself, prove these modelling conventions physically correct. Its equivalence claims below concern the chosen mathematical model. Enabling `use_tight_c13_m` changes the LP relaxation and therefore defines a different bound-comparison protocol, even when its interval constants are valid for the same integer schedules.

The three-day A/B deferral cap and unrestricted `none` option define different $\mathcal{Q}_j$ sets and must be compared separately. Likewise, the compact flag that activates C2–C3 must not be interpreted as adding the heuristic scheduler's explicit ferry events. The proposed initial protocol keeps restrictive station-wait filtering disabled.

Let $\mathcal{R}_j$ index all elements of the finite set $\mathcal{Q}_j$, with $v^r_j$ denoting pattern $r$. A pattern is a complete vector of decisions, not merely a tuple of flight identifiers. Identical flight assignments with different check days or trigger placements must remain available. An idle pattern is admissible only if it satisfies every local constraint, including initial C/D deadlines. A maintenance-only pattern may be used only if the existing trigger formulation actually permits it.

## 7.4 Pattern coefficients and the full master problem

Define the coverage coefficient and full pattern cost by

$$a_{ijr} = x^r_{ij}, \tag{7.8}$$

$$C_{jr} = \sum_{i \in \mathcal{F}:(i,j)\in X} c_{ij} x^r_{ij} + 100 \sum_{d \in \mathcal{D}} \sum_{c \in \mathcal{C}} K^{\text{dur}}_c s^r_{jdc}. \tag{7.9}$$

Here $s_{jdc}$ is the binary maintenance-start variable and $K^{\text{dur}}_c$ is the check duration in days. The compact objective charges 100 times the full check duration for each start. The local start-link rows include $s_{jdc} \le y_{jdc}$ and $s_{jdc} \ge y_{jdc} - y_{j,d-1,c}$ (with $s_{jdc} \ge y_{jdc}$ on the first horizon day). Thus a 0-to-1 transition forces $s = 1$ and an inactive day forces $s = 0$; on a continuing 1-to-1 day either binary value satisfies these rows. Because the objective coefficient is positive, an optimum chooses $s = 0$ on such continuation days. The exact local pattern set retains the original variable and inequalities rather than tightening them. This term is part of the current objective; omitting it would

make the master objective inequivalent even if every feasible pattern were present. The assignment-cost lookup uses the actual input flight row, not an assumption that identifiers are one-based. For any candidate local vector $v_j$, write its complete objective contribution as $f_j(v_j)$; thus $C_{jr} = f_j(v_j^r)$.

Let $\mathcal{I}_{dc} = \{i : d \in D(i,c)\}$ be the trigger flights returned by the implementation's day/check index, and let $\mathcal{F}_m(k,d) = \{i : \text{dest}(i) = k, \text{arrivalDay}(i) \leq d\}$. The exact trigger set for C10 is $\mathcal{T}_{kdc} = \mathcal{I}_{dc} \cap \mathcal{F}_m(k,d)$. Only instantiated aircraft/flight trigger indices are included. For a fixed-tail pattern $r$, the capacity coefficient is

$$b_{kdjr} = \sum_{c \in \mathcal{C}} \sum_{i \in \mathcal{T}_{kdc}} \mathbf{1}\{(i,j,d,c) \in Z\} z^r_{ijdc}. \tag{7.10}$$

This reproduces C10's trigger/start accounting. It does not silently charge capacity on every occupied day of a multi-day check. Local duration blocking continues to be enforced by C14b and the other retained rows. Denote the reference capacity by $K_{kd}$; when the input has a station-wide constant, $K_{kd} = K_k$ for every day.

With $\lambda_{jr} = 1$ selecting pattern $r$ for tail $j$, the full integer master is

$$\min \quad \sum_{j \in \mathcal{P}} \sum_{r \in \mathcal{R}_j} C_{jr} \lambda_{jr} \tag{7.11}$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{P}} \sum_{r \in \mathcal{R}_j} a_{ijr} \lambda_{jr} = 1 \qquad \forall i \in \mathcal{F}, \tag{7.12}$$

$$\sum_{r \in \mathcal{R}_j} \lambda_{jr} = 1 \qquad \forall j \in \mathcal{P}, \tag{7.13}$$

$$\sum_{j \in \mathcal{P}} \sum_{r \in \mathcal{R}_j} b_{kdjr} \lambda_{jr} \leq K_{kd} \qquad \forall k \in \mathcal{K}, d \in \mathcal{D}, \tag{7.14}$$

$$\lambda_{jr} \in \{0,1\} \qquad \forall j \in \mathcal{P}, r \in \mathcal{R}_j. \tag{7.15}$$

The equality in (7.13) ensures a complete local state is selected for every tail, even if its flight assignment is empty.

**Proposition 7.1** (Integer equivalence)**.** *Suppose every local integer pattern of the selected compact model is included in $\mathcal{R}_j$, and C1 and C10 are its only cross-tail constraints. Then the full integer master and that compact model have equal optimal objectives, with cost-preserving feasible-solution mappings in both directions.*

*Proof.* A compact feasible solution determines one $v_j \in \mathcal{Q}_j$ for each tail. Select its indexed column. C1 gives (7.12), C10 gives (7.14), and exactly one pattern is selected per tail. Conversely, binary master variables and the convexity equations select one member of $\mathcal{Q}_j$ per tail. Its local constraints hold, while the two shared master groups reproduce C1 and C10. Equation (7.9) preserves the objective in either mapping. □

The separability premise must be checked against the active implementation options

before this proposition is used computationally. The patterns are not claimed to be physically connected routes beyond what the retained compact routing constraints themselves establish.

## 7.5 Continuous relaxation and bound strength

Replace (7.15) by $\lambda_{jr} \geq 0$. The convexity equations already imply $\lambda_{jr} \leq 1$. Its optimum is denoted $z_{\mathrm{DW}}$. A fractional selection projects into original variables as

$$\bar{v}_j = \sum_{r \in \mathcal{R}_j} v_j^r \lambda_{jr} \in \mathrm{conv}(\mathcal{Q}_j). \tag{7.16}$$

Let $P_j$ be the bound-preserving continuous relaxation of the same local compact block. Because $P_j$ is convex and contains $\mathcal{Q}_j$, $\mathrm{conv}(\mathcal{Q}_j) \subseteq P_j$.

**Proposition 7.2** (Relaxation ordering)**.** *For identical objective, local domains and shared constraints, and finite optima of feasible models,*

$$z_{\text{compact LP}} \leq z_{\mathrm{DW}} \leq z^* \leq U, \tag{7.17}$$

*where $z^*$ is the common integer optimum and $U$ is any validated incumbent.*

*Proof.* Every projected master-LP solution lies in each $P_j$ and satisfies the same shared rows. It is therefore feasible in the compact relaxation, so its minimum cannot be below the compact-LP minimum. Every integer master solution is feasible in the master relaxation, giving the second inequality. The last inequality is the definition of an incumbent upper bound. □

Equality is possible; no strict improvement or runtime advantage follows from this proposition. The comparison uses the standard bound-preserving LP, not a legacy procedure that reopens zero binary bounds. Even a stronger root bound may be expensive to obtain if pricing requires difficult MILP solves.

## 7.6 Restricted master, dual prices and exact pricing

Since the full pattern sets can be enormous, begin with subsets $\widehat{\mathcal{R}}_j \subseteq \mathcal{R}_j$. The restricted master problem (RMP) is the master LP using only these subsets. In this section assume that the RMP is feasible without artificial variables and solved to optimality. Let $\pi_i$ and $\alpha_j$ be unrestricted dual variables for coverage and convexity, respectively,

and let $\beta_{kd} \leq 0$ be the dual variables for capacity. The restricted dual is

$$\max \quad \sum_{i \in \mathcal{F}} \pi_i + \sum_{j \in \mathcal{P}} \alpha_j + \sum_{k \in \mathcal{K}, d \in \mathcal{D}} K_{kd} \beta_{kd} \tag{7.18}$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{F}} a_{ijr} \pi_i + \alpha_j + \sum_{k,d} b_{kdjr} \beta_{kd} \leq C_{jr} \qquad \forall j, r \in \widehat{\mathcal{R}}_j, \tag{7.19}$$

$$\pi_i, \alpha_j \in \mathbb{R}, \quad \beta_{kd} \leq 0. \tag{7.20}$$

The capacity sign is appropriate for a minimization primal with a less-than-or-equal row. Thus $-\beta_{kd} b_{kdjr}$ penalizes use of a scarce station-day resource in the reduced cost.

For each tail, solve the pricing MILP

$$p_j = \min_{v_j \in \mathcal{Q}_j} \quad f_j(v_j) - \sum_{i \in \mathcal{F}} \pi_i x_{ij} - \sum_{k,d} \beta_{kd} b_{kd}(v_j) - \alpha_j. \tag{7.21}$$

Here $b_{kd}(v_j)$ is the linear trigger expression in (7.10). A feasible pricing solution with negative reduced cost defines a column to add; it need not be the pricing optimum to be useful. However, failure of a heuristic to find such a column is not an optimality certificate. Exact termination requires a proven lower bound on the pricing objective for every tail.

The initial pricing design retains the compact local MILP rather than replacing it by a resource-constrained shortest-path algorithm. A future label-setting implementation would need an equivalence proof covering alternative maintenance timings, hierarchy resets and initial conditions. The existing greedy timeline simulator can suggest seed patterns only after they have been checked against $\mathcal{Q}_j$. Its deterministic maintenance placement does not enumerate every feasible column.

## 7.7 Certified lower bounds with incomplete pricing

Restricting columns in a minimization LP can increase its objective. Hence $z_{\text{RMP}} \geq z_{\text{DW}}$, but $z_{\text{RMP}}$ alone need not be a lower bound on $z^*$. A bound must account for columns that remain absent.

Let $\underline{p}_j \leq p_j$ be a certified lower bound from pricing. A pricing incumbent provides an upper bound on $p_j$, not the required lower bound. Define $\delta_j = \min\{0, \underline{p}_j\}$.

**Proposition 7.3** (Pricing-corrected lower bound)**.** *With feasible optimal Phase-II RMP duals and a finite certified $\underline{p}_j$ for every nonempty local pattern set,*

$$L = z_{\text{RMP}} + \sum_{j \in \mathcal{P}} \delta_j \leq z_{\text{DW}} \leq z^*. \tag{7.22}$$

*If every pricing optimum is certified nonnegative, then $z_{\text{RMP}} = z_{\text{DW}}$.*

*Proof.* For every pattern of tail $j$, its reduced cost is at least $p_j \geq \underline{p}_j \geq \delta_j$. Replace $\alpha_j$ by $\alpha'_j = \alpha_j + \delta_j$ while keeping the other dual variables unchanged. All reduced costs become the previous reduced costs minus $\delta_j$, hence are nonnegative. These adjusted duals are feasible for every column of the full master. Their objective is (7.22), proving the bound by weak duality. If the original duals already have no violated column, they are feasible for the full dual; the RMP primal is also full-master feasible, so their equal objectives prove optimality. □

The proof is stated in exact arithmetic. In implementation, solver feasibility and objective-bound tolerances must be recorded and handled conservatively. For example, if only $p_j \geq -\varepsilon$ is certified for every tail, $z_{\mathrm{RMP}} - |\mathcal{P}|\varepsilon$ is the corresponding mathematical lower bound; a zero correction is not justified by that tolerance alone. Unresolved pricing with no valid finite bound yields no new certificate. An inherited node bound can still be retained. A proven empty local set implies node infeasibility, not a numeric pricing correction.

## 7.8 Feasible initialization and Phase I

A small initial pool may not cover all flights or contain a pattern for every tail. A feasible full integer solution is helpful but is not required to initialize column generation. Introduce nonnegative artificial variables $s_i^+, s_i^-$ for coverage and $t_j^+, t_j^-$ for convexity:

$$\sum_{j,r} a_{ijr}\lambda_{jr} + s_i^+ - s_i^- = 1 \qquad \forall i, \tag{7.23}$$

$$\sum_{r} \lambda_{jr} + t_j^+ - t_j^- = 1 \qquad \forall j. \tag{7.24}$$

Retain station capacities and minimize the artificial violation

$$\Phi = \sum_i w_i(s_i^+ + s_i^-) + \sum_j w_j(t_j^+ + t_j^-), \qquad w_i, w_j > 0. \tag{7.25}$$

With nonnegative capacities, the all-zero real-column solution and positive right-hand-side artificials provide an initial feasible Phase-I RMP. Real columns have zero direct cost in Phase I. Their pricing objective uses Phase-I duals and zero pattern cost, not $C_{jr}$ from Phase II.

If the artificial objective reaches zero, fix the artificial variables to zero and optimize the original objective. If exact Phase-I pricing certifies that the complete Phase-I optimum is positive, the full master LP is infeasible, and therefore so is its integer restriction. Positive artificial values in an incompletely priced RMP do not prove infeasibility. Original cost bounds must never be computed from the Phase-I objective.

## 7.9 Branching, incumbents and node bounds

At node $N$, let $\mathcal{Q}_j^N$ contain local patterns consistent with all inherited decisions. Column generation and bound certification are repeated over these node-specific sets. Pricing only at the root is insufficient.

### 7.9.1 Branching in the original variable space

For a fractional master solution, reconstruct all original binaries through (7.16). Prefer an assignment component $0 < \bar{x}_{ij} < 1$ and create the children

$$x_{ij} = 0 \qquad \text{and} \qquad x_{ij} = 1. \tag{7.26}$$

Each child excludes incompatible existing columns and imposes the same fixing in every future pricing solve for tail $j$. Other tail pools are unchanged, although coverage propagation may yield additional valid fixings.

Integral assignments alone do not establish an integral scheduling state. If the assignment components are integral but a maintenance component is fractional, branch on an original binary component $v_{jq}$, for example a $y$, $z$ or hierarchy indicator:

$$\mathcal{Q}_j^{N0} = \{v_j \in \mathcal{Q}_j^N : v_{jq} = 0\}, \qquad \mathcal{Q}_j^{N1} = \{v_j \in \mathcal{Q}_j^N : v_{jq} = 1\}. \tag{7.27}$$

These children partition the original integer patterns. Branching only on an existing $\lambda$ risks allowing future columns to bypass the intended decision, whereas original-variable fixings are directly enforceable in pricing.

If every reconstructed original binary is integral, each projected $\bar{v}_j$ belongs to $\mathcal{Q}_j$: it is an integer vector satisfying the linear local constraints by convexity. Together with the master linking rows it is a compact-model feasible solution, even if some $\lambda$ values remain fractional. Its objective is preserved. The reconstructed certificate must nevertheless be checked against all active original constraints before it updates the incumbent $U$.

### 7.9.2 Pruning and global certification

A node is closed only by a certified infeasibility result, a valid bound $L_N \geq U$, or a certified node-optimal integral projection. Discovering an integral incumbent during incompletely priced optimization does not alone close the node. A restricted integer master may improve $U$ if its selected patterns satisfy the original model, but its reported restricted-pool optimality is not full-problem optimality.

Children inherit the parent's valid lower bound. When new pricing certificates are available, use the larger of that inherited bound and the new valid correction. For the

open-node set $\mathcal{O}$ and a finite incumbent, a global lower bound is

$$L_{\text{global}} = \min\left\{U, \min_{N\in\mathcal{O}} L_N\right\}. \tag{7.28}$$

When no nodes remain, the inner minimum is $+\infty$; a finite incumbent is then optimal if every closure was justified. Without an incumbent, exhausting all nodes by certified infeasibility proves infeasibility. At interruption, open nodes lacking a new pricing certificate remain represented by their inherited valid bound, potentially $-\infty$.

For finite bounds and nonzero incumbent cost, report

$$g = 100\,\frac{\max\{0, U - L_{\text{global}}\}}{|U|}. \tag{7.29}$$

Zero-cost and missing-bound cases require explicit handling rather than a fabricated finite percentage. All solver calls share a global remaining-time budget; separately granting the full budget to every pricing call would not constitute a fixed-budget comparison.

## 7.10 Proposed algorithm

Algorithm 1 describes the intended controller. In exact arithmetic, finite binary local sets, complete pricing and original-binary branching give a finite exact search. This is not a practical runtime guarantee. Degenerate master LPs, expensive pricing and duplicate columns can make convergence slow. Dual stabilization or heuristic pricing may be useful implementation options, but certification must use valid duals and complete pricing bounds, not an unverified stabilized objective.

## 7.11 Illustration of the restricted-master bound issue

The following is an algebraic illustration, not an experiment on the ABCD benchmark. Consider one flight and two tails. Both admit an idle pattern. The flight costs 100 on tail 1 and 10 on tail 2, and no coupling resource is binding. Suppose the initial pool contains the flight pattern only for tail 1, together with both idle patterns. The RMP assigns the flight to tail 1 and has objective 100, but the actual integer optimum is 10. Thus reporting 100 as a lower bound would be false.

An optimal RMP dual solution is $\pi = 100$ and $\alpha_1 = \alpha_2 = 0$. The missing tail-2 flight pattern has reduced cost $10 - 100 = -90$. With pricing minima $p_1 = 0$ and $p_2 = -90$,

**Algorithm 1** Proposed full-ABCD branch-and-price controller

Initialize root, validated seed patterns, $U \leftarrow +\infty$ and open-node queue
**while** the queue is nonempty and global time remains **do**
    Select node $N$; filter pools and pricing models by inherited fixings
    Run Phase-I column generation when a feasible master is unavailable
    **if** full Phase-I infeasibility is certified **then**
        Close $N$ and continue
    **end if**
    **if** Phase I remains unresolved **then**
        Retain $N$ with its inherited bound and mark it unfinished
        Continue to another node, or stop if the global budget is exhausted
    **end if**
    **repeat**
        Solve the Phase-II RMP LP and obtain consistent dual prices
        Price every tail over $\mathcal{Q}_j^N$ within the remaining budget
        Add new negative-reduced-cost patterns and retain pricing certificates
        Update $L_N$ from the pricing correction when its premises hold
    **until** full LP certification or interruption of column generation
    Validate any reconstructed integer schedule; update $U$ if improved
    **if** $N$ is certified infeasible, bound-prunable, or node-optimal integral **then**
        Close $N$
    **else if** the full node LP is certified and its original projection is fractional **then**
        Branch on an original binary and enqueue both children
    **else**
        Retain $N$ as unfinished with a valid bound
    **end if**
**end while**
Report incumbent, global bound, status and complete timing/provenance

the correction yields

$$L = 100 + \min\{0, 0\} + \min\{0, -90\} = 10. \tag{7.30}$$

Adding that pattern lets the master reach 10. In the integrated model, station-capacity duals and alternative maintenance placements perform the same coordinating role, but with much larger local pattern spaces.

## 7.12 Validation and planned computational evaluation

Before implementation results are admitted to the thesis, the following gates are required. First, tiny exhaustive local-pattern enumeration must establish that the priced sets reproduce the selected compact constraints and bounds. The complete enumerated master LP must match certified column generation; its integer optimum and branch-and-price must match the compact MILP. Second, targeted cases must cover shared station bottlenecks, initial A/B exposure, C/D deadlines, multi-day checks, hierarchy substitution, alternative maintenance placements, infeasible idle patterns and sparse trigger domains. Third, branching tests must include integral assignments with fractional maintenance states, while interruption tests must distinguish feasible pricing patterns from certified pricing lower bounds.

Matched experiments should then compare the bound-preserving compact LP, the certified root Dantzig–Wolfe relaxation, the selected strengthened compact MILP and the full branch-and-price search. Use the same canonical input, ABCD checks, cap, cost, station accounting, solver version and thread policy. Capped and unrestricted studies remain separate protocols. Start with small verified cases before extending the grid to 7-, 14-, 21- and 30-day horizons.

Record the certified lower bound, incumbent, residual gap, proof status, generated columns, pricing iterations and calls, search-tree nodes, master and pricing time, construction time and total wall time. Source and input hashes and all model options are required. Report bound improvement together with its computational cost; a stronger root bound is not sufficient evidence of a faster exact algorithm. Memory reductions require actual measurements, not an inference from the number of master variables alone.

The existing route experiment enumerates routing-only paths and does not implement pricing or maintenance-aware branch-and-price. Its results therefore cannot be relabelled as validation of this chapter. Similarly, the 30-day A-only Gantt illustrates a different solved configuration, not a full-ABCD branch-and-price schedule.

## 7.13 Review decisions before coding

The proposed mathematical design should be evaluated on four questions: whether the local/shared decomposition reproduces every active compact row; whether C10 trigger accounting is the intended comparison resource; whether the bound and Phase-I certificates are sufficiently explicit; and whether exact MILP pricing is an acceptable first implementation despite its potential cost. Approval of the chapter does not itself establish computational effectiveness. Implementation and a reproducible experimental study remain subsequent tasks, and their results must determine the final scope of the thesis's contribution claims.

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